

Code-Layout, Readability and Reusability

Date	06 July 2026
Team ID	
Project Name	Ledgerline – Credit Card Approval Prediction System
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing used throughout the code	Yes	Standard 4-space Python indentation throughout
2	Proper File Structure	Files and folders are logically organized	Yes	Separate data/, models/, templates/, static/ folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. FEATURE_COLUMNS, DECISION_THRESHOLD, scale_pos_weight
4	Function / Method Names	Functions are descriptively named	Yes	e.g. build_feature_row(), predict_row()
5	Code Comments	Inline and block comments explain logic	Yes	Section banners + inline notes on non-obvious steps (e.g. the DAYS_EMPLOYED sentinel value)
6	Modular Design	Code is split into reusable functions/modules	Yes	Data generation, feature engineering, and training are separate scripts
7	No Redundant Code	Duplicate or unused code is removed	Yes	One shared preprocessing path used by both single and batch prediction
8	Error Handling	Exceptions and errors are	Yes	try/except around

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
		handled gracefully		prediction and batch file parsing, with user-facing error messages

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High/Medium/Low)
1	build_feature_row()	Python	Used by both the single-applicant form and the batch CSV screener	High
2	predict_row()	Python	Used by /predict, /api/predict, and /batch/predict	High
3	feature_columns.json	JSON	Shared between training pipeline and Flask app	High
4	common.js-style Flask templates (base.html)	Jinja2	Extended by all 4 pages (index, result, batch, about)	High
5	train_models.py model-comparison loop	Python	Generic over any scikit-learn-compatible classifier	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation between data/model code and the web app
Readability	5	Descriptive names and comments throughout
Reusability	4	Core preprocessing/prediction logic is shared; could be split into a separate package for reuse outside this app
Documentation / Comments	5	README, inline comments, and a full Jupyter notebook walkthrough
Overall Score	4.75	Strong overall code quality for a project of this scope